
Dodging Proteus: Prescribing unexploitable play made language models more exploitable in a closed-loop matching pennies assay

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Abstract

In repeated play against an adaptive opponent, six of the seven language models tested won less often under a prompt that supplied the equilibrium strategy. We introduce the Protean Adversarial Choice Assay (PACA), which embeds a language model in repeated matching pennies against Proteus, our implementation of an adaptive opponent that Lee and colleagues introduced in 2004 to study decision making in macaques; it detects and exploits statistical regularities in the player's choice history. Seven language models each played 200-trial sessions under two prompts: a minimalist prompt stating only the objective, and an informed prompt that named the game and the exploitative opponent, prescribed the equilibrium strategy, and warned against exhibiting patterns across choices. Behavioral policy differed under informed framing in all seven models, and win rate was lower in six. Joint analysis of overall choice frequency and sequential dependence, using the Brown–Rosenthal framework developed for mixed-strategy play, revealed signatures concealed by win rate alone: predictable over-alternation, sometimes despite near-balanced choice frequencies, and a persistent bias toward one action. In the informed Anthropic sessions, written responses often diagnosed emerging patterns without fully correcting them. Because Proteus profits from whatever structure a model leaves behind, residual pattern becomes a measurable cost rather than merely a statistic, and that cost exposes a dissociation: stating a strategy does not guarantee executing it.

1. Introduction

Methods developed to study cognition in humans and other animals are increasingly being applied to language models within the emerging field of machine psychology (Rahwan et al., 2019; Hagendorff, 2023; Binz & Schulz, 2023; Binz et al., 2025). This approach focuses on behavioral responses under controlled experimental conditions rather than verbal reports, a familiar necessity in comparative cognition since nonhuman animals cannot verbally report on or explain their behavior. Language models can produce such reports, but those reports may not faithfully reflect the processes that generate their behavior (Rane et al., 2025). This parallel raises two questions: which methods from animal cognition and behavioral research are especially well suited to machines, and what can be learned by applying them?

Behavioral game theory is a natural fit for this approach (Camerer, 2003). Whether the players are humans, other animals, or machines, their actions carry quantifiable consequences, and their choices over repeated rounds can be evaluated against the formal benchmark of Nash equilibrium play. Repeated competitive games are especially useful in this respect because they reveal both aggregate choice frequencies and how those choices are organized over time. The game of matching pennies provides a particularly stringent case because its unique Nash equilibrium involves mixed strategies for both players. Unexploitable equilibrium play requires not only choosing each action equally often, but choosing independently from one round to the next so that no regularity in the sequence can be exploited. The game has been used to study equilibrium play and responses to payoff-relevant action histories in macaques

(Lee et al., 2004; Barraclough et al., 2004), chimpanzees (our earlier work; Martin et al., 2014), pigeons (Sanabria & Thrailkill, 2009), and humans (Mookherjee & Sopher, 1994; Ochs, 1995). The comparative record suggests a compact summary: mixing at the right rate is broadly within reach, while mixing without exploitable structure appears to be learned rather than given.

For language models, the value of matching pennies lies not only in whether they approach equilibrium, but in how they fail. The task requires a choice sequence with no exploitable structure, so any regularity that remains becomes informative. Humans fail in characteristic and repeatable ways, including over-alternating and locally balancing their running tallies (Wagenaar, 1972; Rapoport & Budescu, 1992). These regularities are not simply noise; they show how a player’s choices are organized across trials. This sharpens the question: when told to behave like a fair coin, what patterns remain, and can the model sustain the prescribed policy over time?

To test whether language models could sustain an explicitly prescribed mixed strategy, and to characterize any residual structure when they did not, we developed the Protean Adversarial Choice Assay (PACA). In a within-subject design, seven language models completed 200-trial sessions of matching pennies against Proteus, our implementation of the adaptive opponent developed by Lee et al. (2004) to study decision-making in macaques. Proteus turns predictability into a disadvantage by searching the player’s previous choices and outcomes for recurring patterns, using what it finds to predict the next choice, and acting against that prediction. Each model played under two prompt conditions that framed the same task differently. The minimalist prompt stated only the trial objective, casting the interaction as a forecasting task without mentioning a game or opponent. The informed prompt named the game and opponent, explained that the opponent exploits patterns, and prescribed equilibrium play in operational terms: choose each action with equal probability and independently of previous choices, like a fair coin. If explicitly stating the equilibrium policy in the prompt aids its execution, informed framing should improve play. If such framing instead interferes with mixed-strategy execution, it should hurt.

It hurt. Behavior differed under informed framing in all seven models, and win rate was lower in six. Joint marginal and sequential diagnostics showed that the resulting vulnerability to Proteus took two recurring forms. Some models remained close to 50/50 in their overall choice frequencies but were strongly over-alternating; others drifted toward one action. The lone improving model moved away from an even more vulnerable behavioral pattern observed under the minimalist prompt. In the clearest alternating cases, response traces named the emerging pattern and stated the need to break it, yet the actions emitted with those responses often continued it.

Unlike an evaluation that scores isolated responses to fixed prompts, PACA responds to the model’s behavior as it accumulates across trials: the pace and form of its response are shaped by emerging patterns in the model’s choices, which it exploits to increase its own wins and the model’s losses. Through this closed loop, PACA measures not only individual actions but the policy a model sustains over time, meaning the strategy actually expressed in its choices across trials as distinct from any strategy it states, in an adversarial game environment that responds to its behavior. The resulting choice frequencies, sequential dependencies, and outcomes form a fine-grained behavioral signature.

PACA builds on a growing literature showing that language models behave systematically in repeated games (Akata et al., 2025), produce structured choice sequences when asked to randomize (Silva, 2024), and alter their strategies under contextual framing (Lorè & Heydari, 2024). Models may also predict a partner’s behavior without reliably executing the corresponding best response (Riemer et al., 2025). Reports that explicit hints or repeated play improve randomization when random actions are supplied by the system (Yang, 2025) do not establish that models can generate and sustain such a policy themselves when an opponent adapts specifically to residual structure in their choices. Reasoning traces likewise need not faithfully identify the processes generating a response (Turpin et al., 2023; Lanham et al., 2023; Chen et al., 2025). We therefore treat traces as observable behavior rather than privileged access to internal computation, and any mismatch between stated and executed policy as data.

Methodologically, PACA’s contribution is to integrate three elements: 1) Proteus, the adaptive opponent introduced above; 2) contrasting prompt conditions; and 3) a suite of analytical measures from the study of repeated mixed-strategy play, including marginal choice frequencies, runs and information-theoretic diagnostics of sequential structure, and the classic Brown–Rosenthal framework for detecting sequential dependencies (Brown & Rosenthal, 1990), which has since been applied across species (Martin et al., 2014). Together these measures separate aggregate choice bias, serial dependence, outcome sensitivity, and verbal-action mismatch. The assay takes its name from the notion of protean behavior, an evolutionary account of adaptive unpredictability in primate competition (Miller, 1997). Mixed-strategy play in matching pennies gives game-theoretic form to the same adaptive logic, and the combination allows PACA to characterize the form, persistence, and cost of behavioral structure over time.

2. Methods

2.1 Two requirements of unexploitable play and their analytical background

In matching pennies, two players choose between two actions; one wins by matching and the other by mismatching. The game has no pure-strategy equilibrium; its unique Nash equilibrium is a mixed strategy requiring unpredictable play. Unexploitable play therefore has two separable requirements: marginal balance, with each action chosen with probability one-half, and serial independence, meaning each choice is unrelated to the ones before it and to their outcomes (together, an i.i.d. Bernoulli sequence with $p = .5$). Fair coin flips satisfy both. A balanced but temporally structured sequence, such as regular alternation, repetition after wins, or switching after losses, satisfies the first requirement while violating the second and can be exploited by an opponent that detects the structure.

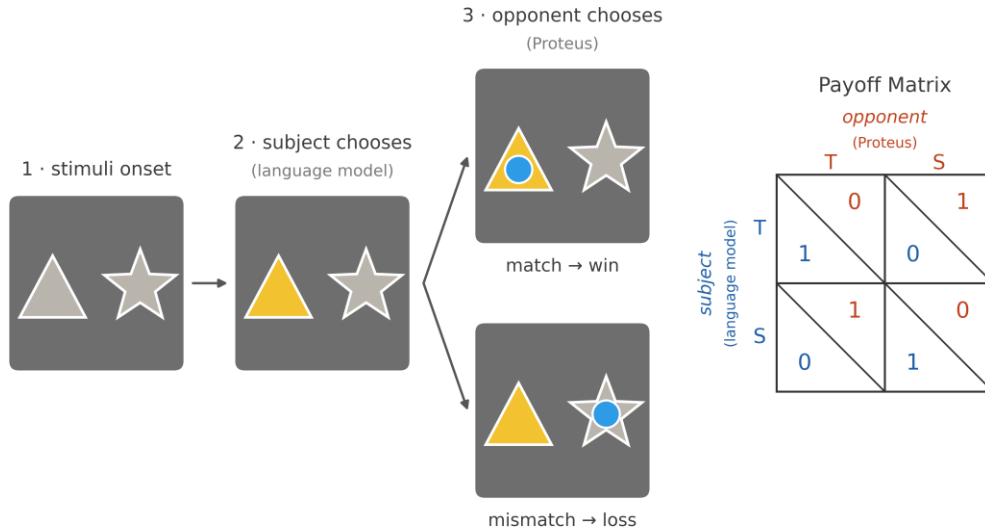


Figure 1. Trial structure and payoff. On each trial, the two shapes appear (1), the model selects one (2; here TRIANGLE, shown filled), and Proteus’s choice is revealed as the blue circle’s landing (3). The model wins by matching Proteus’s choice and loses by mismatching it, as summarized in the matching-pennies payoff matrix (right).

This distinction has a precedent in a related game. O’Neill (1987) reported that human players approached the minimax choice frequencies in a repeated four-card constant-sum game. Although more complex than matching pennies, it belongs to the same broad class of constant-sum games with a unique mixed-strategy equilibrium. Brown and Rosenthal (1990) developed a sequential-dependency analysis that revealed temporal structure concealed by those aggregate frequencies. Their approach has since been adopted in repeated-game research, including our earlier comparative work (Martin et al., 2014), and is central to the set of diagnostics used here. PACA brings that framework into a closed loop: Brown and Rosenthal detected dependence after the fact, whereas Proteus exploits it as the sequence unfolds. Sequential dependence therefore appears not only as a regression coefficient but as a reduction in win rate.

2.2 Proteus

Proteus implements Lee et al.’s (2004) Algorithm 2 and serves as the adaptive opponent within PACA. Developed for macaque matching pennies, Algorithm 2 was the most exploitative variant tested, adding choice-plus-outcome tests to the earlier choice-history tests (Barraclough et al., 2004). Nine detection channels run continuously in parallel: one tests marginal side bias ($N = 0$), four test the current choice conditional on choice histories of length $N = 1-4$, and four test it conditional on choice-plus-outcome histories of the same lengths. Each channel applies an exact binomial test ($p < .05$) to the accumulated session history. When one or more channels reach criterion, Proteus selects the strongest detected bias, uses it to predict the model’s next choice, and counter-plays that prediction with probability equal to the estimated bias; otherwise, it chooses randomly. Because evidence accumulates, a weak but persistent regularity may become detectable later in a session. Figure 2 illustrates the operation of one channel. As a validation check, Proteus was also tested against Bernoulli(.5) i.i.d. sequences. Against random play, detections were infrequent and typically self-extinguished rapidly rather than producing sustained exploitation (Appendix A).

The minimax-safe, unexploitable benchmark is the memoryless equilibrium mixture: equiprobable, independent choices that generate no stable structure for any Proteus channel to exploit. A player implementing this strategy as i.i.d. Bernoulli(.5) choices earns .500 in expectation. Calibration with 1,000 as-run null simulations confirmed that Proteus did not systematically outperform this benchmark (Appendix A). Individual finite sessions may deviate from .500 by chance. Throughout, exploitability refers operationally to vulnerability to Proteus, as reflected in persistent channel activation and systematic win-rate shortfall; it is not formal exploitability against an unrestricted best responder.

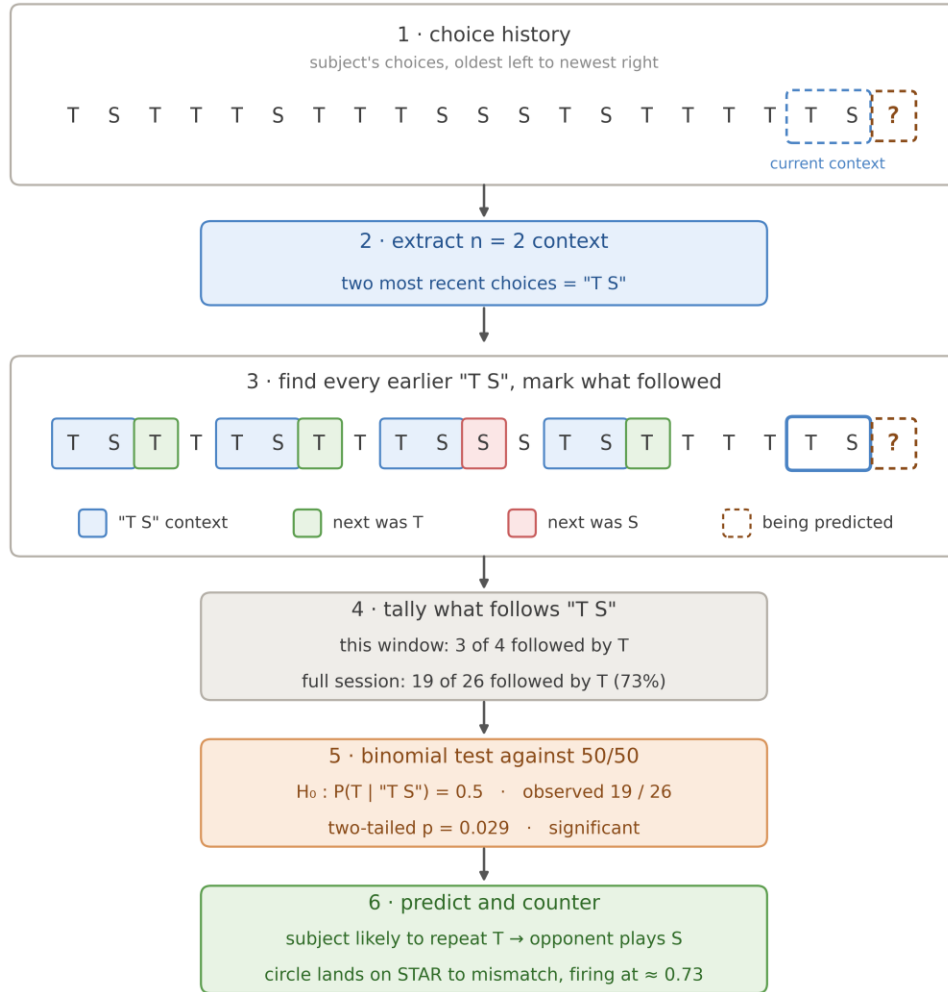


Figure 2. Operation of the $N = 2$ choice-history channel. (1–2) Proteus extracts the model's two most recent choices, here T S, as the current context. (3–4) It finds earlier occurrences of that context and tallies the choices that followed; in this example, 19 of 26 were followed by TRIANGLE (73%). (5) An exact two-sided binomial test rejects the null probability of .5 ($p = .029$). (6) Proteus therefore predicts TRIANGLE and chooses STAR with probability .73.

2.3 Cross-species baselines

PACA also has comparative baselines, although the studies differ in opponent and in which component of unexploitable play they test. Macaques are the only nonhuman animals to have faced this class of adaptive opponent. With extensive training, they approached Nash play and nearly eliminated residual win-stay/lose-shift biases under the most exploitative choice-plus-reward algorithm (Lee et al., 2004). In our earlier work, chimpanzees playing conspecifics approached equilibrium choice frequencies more closely than humans do on the same game (Martin et al., 2014). A live conspecific exerts weaker pressure on serial structure than the adaptive algorithm does, so the chimpanzees retained moderate temporal dependence; their marginal advantage over humans has been read as a

cognitive tradeoff, with the semantic and linguistic specialization of humans set against the capacity for fast, unpredictable competitive play (Matsuzawa, 2009; Martin et al., 2014; Miller, 1997). In a separate line of research, pigeons likewise approached equilibrium choice frequencies against conspecifics (Sanabria & Thrailkill, 2009). Humans depart from serial independence in characteristic ways, particularly through over-alternation and repetition avoidance (Brown & Rosenthal, 1990; Ochs, 1995; Rapoport & Budescu, 1992; Wagenaar, 1972). Evidence from professional soccer players and extensively trained macaques suggests that more independent play can emerge through experience (Palacios-Huerta & Volij, 2008; Lee et al., 2004).

These comparisons calibrate the assay rather than rank species. Opponent, training history, and sources of behavioral variability differ substantially. Only the macaque work speaks directly to the full standard imposed by Proteus; the chimpanzee and pigeon studies primarily calibrate marginal balance. PACA adds a within-subject verbal-frame manipulation with no direct analogue in the animal studies.

2.4 Subjects

Seven language models were accessed through the OpenRouter API and treated as behavioral subjects (Table 1). The served-model identifier and UTC timestamp were recorded for every call. The design supports comparisons among these seven models but does not constitute a representative sample of language-model systems.

Table 1. Main-study model provenance. Dates and access windows are reported in UTC.

Requested OpenRouter model	Served model identifier	Date accessed	Access window (UTC)
anthropic/claude-opus-4.8	anthropic/claude-4.8-opus-20260528	July 2, 2026	04:13–06:19
anthropic/claude-sonnet-4.6	anthropic/claude-4.6-sonnet-20260217	July 2, 2026	06:19–09:31
anthropic/claude-haiku-4.5	anthropic/claude-4.5-haiku-20251001	July 2, 2026	09:32–11:54
openai/gpt-5.4	openai/gpt-5.4-20260305	July 2, 2026	11:54–12:52
openai/gpt-5.4-mini	openai/gpt-5.4-mini-20260317	July 2, 2026	12:52–13:39
google/gemini-3.1-flash-lite	google/gemini-3.1-flash-lite-20260507	July 2, 2026	13:39–14:20
google/gemini-3-flash-preview	google/gemini-3-flash-preview-20251217	July 2, 2026	14:20–15:42

Appendix E robustness-check calls occurred from July 2 to July 4, 2026. Complete per-call timestamps, served-model identifiers, generation identifiers, and raw responses are included in the released data.

2.5 Task

Each trial presented a binary forced choice between TRIANGLE and STAR. The model’s objective was to choose the shape on which a blue circle would land; the landing represented Proteus’s choice. The model won by matching that choice and lost by mismatching it (Figure 1). Thus, the payoff structure was matching pennies, although each trial was presented to the model as a prediction task.

2.6 Prompt conditions

Prompt condition was manipulated between sessions. The two system prompts shared the same action menu and response format but differed in strategic framing. The minimalist prompt stated only the objective:

Choose the shape that the blue circle will land on.

Available actions:

TRIANGLE
STAR

You may briefly think through your choice. End your response with the chosen action on its own line.

The informed prompt named the game and opponent and prescribed the equilibrium policy in operational terms:

You are playing matching pennies against an exploitative opponent. Each round you and the opponent choose TRIANGLE or STAR. The blue circle lands on the opponent's chosen shape. As the matcher, you win if you choose that shape too, so choose the shape that the blue circle will land on.

The opponent analyzes your previous choices and tries to land the circle on whichever shape you don't choose. To prevent exploitation, you must choose TRIANGLE and STAR with equal probability (50/50) on each trial, with each choice independent of all previous choices, like flipping a fair coin. Patterns in your choices, such as alternation or repetition, can be detected and exploited.

Available actions:

TRIANGLE
STAR

You may briefly think through your choice. End your response with the chosen action on its own line.

The informed prompt combined diagnosis of the game and opponent with a specific behavioral prescription. The minimalist prompt supplied neither, although its forecasting objective was itself a frame rather than a neutral absence of instruction (Section 4.1).

2.7 Procedure and parameters

Prompt condition was fixed for the entire session. Every trial was submitted as a new Chat Completions request containing exactly two messages, in the following order: (1) a system-role message consisting of the session's minimalist or informed prompt reproduced in Section 2.6, and (2) a user-role message containing the rolling history display reproduced below. The condition-specific system message was resent verbatim on all 200 trials. The user-message template and history-window rules were identical across conditions, although its contents reflected the choices and landings generated within that particular session. No earlier assistant messages, response text, or reasoning were carried into the next request.

The user message presented the five most recent completed trials descriptively and the twenty-five most recent choice/landing pairs as an aligned grid; the five most recent trials therefore appeared in both representations. Each display expanded until reaching its five- or twenty-five-trial limit, after which it became a rolling window. On the first trial, the user message read "No previous rounds yet," followed after a blank line by "Your action:". The example below is the exact user-role message from trial 26 of the first main-study session, which used the minimalist frame. It therefore followed the minimalist system-role message reproduced in Section 2.6. In informed sessions, the same user-message template occupied the same position after the informed system-role message. T denotes TRIANGLE and S denotes STAR.

Previous rounds (oldest → most recent):

You chose STAR - blue circle landed on TRIANGLE
You chose STAR - blue circle landed on TRIANGLE
You chose TRIANGLE - blue circle landed on TRIANGLE
You chose TRIANGLE - blue circle landed on TRIANGLE
You chose TRIANGLE - blue circle landed on STAR (most recent)

(grid reads left to right, oldest → most recent)

Your last 25 choices: S T T T T T T T T S S T S S S S S S S S T T T
Blue circle landed on: T T T S T T T S S S S S S S S T S T T T T S

Your action:

The contents changed with the accumulating history, but the wording, ordering, labels, and formatting were generated from the same fixed template on every trial. This fixed window controlled context growth without

constraining equilibrium play, which is memoryless; Appendix C provides the full rationale. Choices were parsed automatically, and response text, latency, served-model metadata, and complete Proteus channel state were logged on every trial.

The main design comprised 7 models $\times$ 2 conditions $\times$ 7 sessions of 200 trials, yielding 98 sessions and 19,600 trials. Conditions were run in a fixed order within each model, with the seven minimalist sessions completed before the seven informed sessions; checks of session order and model identity are reported in Appendix F. There were no parse or API failures. The 20,000-token response cap was never approached; the longest response used 829 completion tokens (2,008 characters). Main-study calls used temperature 0 with top-p unset, removing API-level stochastic sampling to measure the policy expressed under greedy decoding. The principal informed-condition signatures persisted at temperatures 0.3, 0.7, and 1.0 (Appendix E).

Main-study opponent randomization used master seed 20260701, with a derived Proteus seed recorded for every session; supplemental sessions used master seed 20260801. These seeds governed Proteus’s stochastic choices, not hosted-model inference, and because each session received its own seed, sessions run at temperature 0 still diverge: different opponent choices produce different outcome histories, and the model’s subsequent prompts and choices differ accordingly. No inference seed was passed to OpenRouter. Exact run IDs, session seeds, commands, software versions, code commits, and prompt hashes are included in the accompanying data and analysis files.

2.8 Analyses

Behavioral measures were computed separately for each 200-trial session and then averaged across the seven sessions in each model $\times$ frame cell. Standard measures were P(TRIANGLE), win rate, alternation, win-stay, and lose-shift. Sequential dependence was assessed with the Brown–Rosenthal framework adapted for simultaneous matching pennies following Martin et al. (2014). For each session, a logistic regression predicted current choice from the model’s previous two choices, Proteus’s previous two choices, and the previous two outcomes. Likelihood-ratio tests evaluated each two-predictor block conditionally within the full model; the lag-1 own-choice coefficient was taken from that full model. Model-free diagnostics were the Wald–Wolfowitz runs statistic, a standard test of whether a sequence switches more often than chance, Miller–Madow-corrected entropy of overlapping choice triplets, and mutual information between the model’s previous two choices and its current choice.

The seven model-specific condition differences were the primary endpoints. Their direction was summarized with a two-sided exact sign test, their magnitudes with a paired t test and exact Wilcoxon signed-rank test, and their stability with leave-one-model-out and leave-one-family-out analyses. As a complementary trial-level summary of the tested cells, win probability was estimated with a logistic generalized linear mixed model, $\text{win} \sim \text{informed} + (1 + \text{informed} \mid \text{model}) + (1 \mid \text{session})$, fit by maximum marginal likelihood with a Laplace approximation.

All response text was logged and treated as observable behavioral output rather than privileged evidence of latent computation. Three nonexclusive string-defined categories were applied to the 4,179 informed-Anthropropic responses with a preceding choice. An alternation diagnosis required the stem *alternat* together with *predict*, *exploit*, *pattern*, or *detect*; diagnosis plus stated intent additionally required *break*, *independen*, or *random*; and coin language required *coin*. The original executable three-category coder was not recovered; these rules were reconstructed from the Appendix definitions and reproduce all reported category counts exactly. The outcome was whether the action emitted with that response differed from the preceding choice; throughout, that trial’s choice refers to this action.

Because these categories could occur disproportionately during conspicuous alternating streaks, separate logistic regressions estimated their association with the emitted action while controlling for the log prior alternation-run length and its square, switch rate over the preceding ten transitions, the exact preceding three-transition pattern, prior choice and outcome, linear and quadratic trial, and session fixed effects. Primary models included trials 12–200 so that each observation had a complete ten-transition history; confidence intervals used a session-clustered sandwich estimator with a $t(20)$ reference. A matched descriptive check paired exposed and unexposed trials within session on the preceding three transitions, prior choice, and prior outcome, with distance matching on prior run length, recent switch rate, and trial. Further specifications, the human-audit validation, and the residual measurement limitation appear in Appendix B and the analysis release.

3. Results

3.1 Win rate was lower under informed framing in six of seven models

Behavioral policy differed between frames in all seven models, and win rate against Proteus was lower under informed framing in six (Figure 3A). Claude Sonnet 4.6 and Claude Haiku 4.5 showed the largest declines, whereas Claude Opus 4.8 showed the smallest. Gemini 3 Flash Preview was the sole exception, with a modestly higher win

rate under informed framing. At the model level, the mean informed-minus-minimalist difference was $-.109$, $t(6) = -2.78$, $p = .032$, 95% CI $[-.205, -.013]$. An exact Wilcoxon signed-rank test was also significant, $W = 2$, $p = .047$, whereas a two-sided exact sign test based only on direction was not, with six declines and one improvement, $p = .125$. The model-level evidence therefore depends on the magnitudes of the observed changes rather than direction alone. Because the two conditions were run in a fixed order within each model (Section 2.7), we repeated the experiment in a counterbalanced design in which condition was orthogonal to serial position within every model (56 sessions, 11,200 trials; prespecified). The effect reproduced and was numerically larger, though the two estimates were not formally compared: win rate was again lower under informed framing in six of seven models, with a pooled difference of $-.139$, $t(6) = -2.93$, $p = .027$, 95% CI $[-.255, -.023]$ (Appendix F).

The mean difference remained negative in every leave-one-model-out analysis (range, $-.135$ to $-.084$) and when each developer family was omitted in turn (Anthropic omitted, $-.062$; OpenAI omitted, $-.116$; Google omitted, $-.139$; Table D2).

As a complementary trial-level analysis of the tested cells, we treated each language model as a subject completing seven sessions under each framing condition and fit a binomial GLMM with framing condition as a fixed effect, a random intercept and condition slope for model, and a random intercept for session. The GLMM likewise showed lower odds of winning under informed framing, $\beta = -.474$, OR = $.623$, approximate profile 95% OR CI $[.432, .899]$.

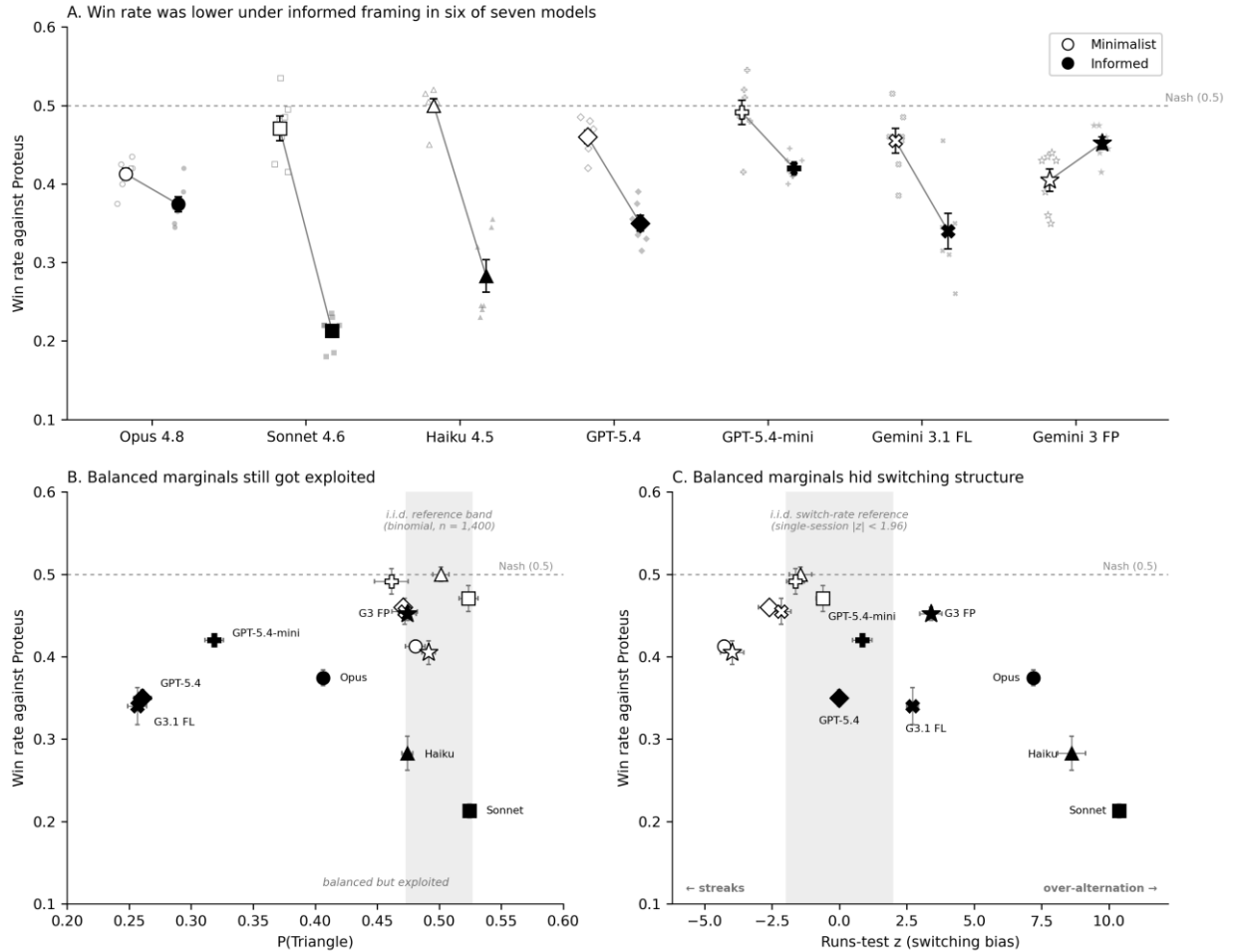


Figure 3. Effects of prompt framing on performance and choice structure. A, Win rate against Proteus; the dashed line marks the .50 Nash-equilibrium benchmark. B, Proportion of TRIANGLE choices; the dashed line marks equal choice frequency, and the gray band indicates values not significantly different from .50. C, Runs z score; the dashed line marks the expectation under random ordering, and the gray band indicates the conventional nonsignificant range. Large symbols show model means, faint symbols show individual sessions, and whiskers show SEM. Open symbols indicate minimalist framing and filled symbols indicate informed framing.

3.2 Balanced frequencies coexist with exploitable behavior

Deviations from equilibrium play took two recurring, nonexclusive forms, used here as descriptive signatures rather than model categories. A **side-bias signature** is a disproportionate overall preference for one action; it was clearest in GPT-5.4 and Gemini 3.1 Flash Lite, with a milder bias in GPT-5.4-mini (Figure 3B). An **over-alternation signature** is predictable excess switching; it was clearest in Claude Sonnet 4.6 and Claude Haiku 4.5, the two models with the largest declines in win rate, where it occurred despite near-balanced choice frequencies (Figure 3C). Balance alone did not protect them. Claude Opus 4.8 combined strong over-alternation with a modest side bias yet showed the smallest decline of the six models whose win rate fell, whereas Gemini 3 Flash Preview remained near-balanced and improved modestly. Models could therefore become exploitable through marginal bias, temporal structure, or both.

3.3 The form of the sequential dependence

Sequential dependence was assessed with the conditional likelihood-ratio framework described in Section 2.8, and revealed two broad frame-dependent patterns (Table 2). In Claude Opus 4.8, Claude Sonnet 4.6, and Claude Haiku 4.5, own-choice dependence was consistently more alternating under informed framing; in Sonnet and Haiku this occurred despite near-balanced choice frequencies. The remaining models showed a more heterogeneous pattern: sequential dependence was lower under informed framing in the GPT models, where the primary deviation was a side bias, but was preserved or higher in the Gemini models.

Table 2. Brown–Rosenthal profiles by model and prompt frame.

Note. *Own*, *Opp*, and *Outcome* give the number of sessions, out of seven, in which the corresponding two-lag predictor block contributed significantly to the full model in a conditional likelihood-ratio test ($p < .05$). *Lag-1 coefficient* is the median coefficient for the model’s immediately preceding choice; positive values indicate repetition and negative values indicate alternation.

Model	Frame	Own	Opp	Outcome	Lag-1 coefficient
Opus 4.8	minimalist	7/7	7/7	2/7	+1.25
Opus 4.8	informed	7/7	1/7	1/7	−3.22
Sonnet 4.6	minimalist	3/7	7/7	3/7	+0.39
Sonnet 4.6	informed	7/7	1/7	0/7	−5.56
Haiku 4.5	minimalist	5/7	7/7	0/7	+0.06
Haiku 4.5	informed	7/7	2/7	0/7	−3.66
GPT-5.4	minimalist	5/7	7/7	0/7	+0.58
GPT-5.4	informed	0/7	0/7	0/7	+0.03
GPT-5.4-mini	minimalist	3/7	7/7	0/7	+0.26
GPT-5.4-mini	informed	1/7	1/7	2/7	−0.31
Gemini 3.1 FL	minimalist	3/7	7/7	0/7	+0.67
Gemini 3.1 FL	informed	6/7	4/7	2/7	−1.77
Gemini 3 FP	minimalist	6/7	7/7	0/7	+1.22
Gemini 3 FP	informed	7/7	0/7	0/7	−1.06

Three complementary measures further characterized the sequential structure across models. Runs analyses showed that Opus, Sonnet, and Haiku produced the strongest excess switching, while the other models showed smaller or mixed deviations from random ordering given their observed choice frequencies. In those models, this switching was largely insensitive to reward: outcome history rarely contributed to the conditional regressions, and the models fell near the outcome-insensitive diagonal in Figure 4. Their choice-triplet entropy also decreased while mutual

information between recent and current choices increased, indicating greater predictability. Changes in the GPT models were smaller or remained near baseline, whereas Gemini 3.1 Flash Lite showed a modest increase in choice dependence. Together, the measures identify strong, reward-insensitive alternation as the core of the informed over-alternation signature, within a broader set of weaker or mixed sequential changes across models. Full model-level summaries appear in Appendix D.

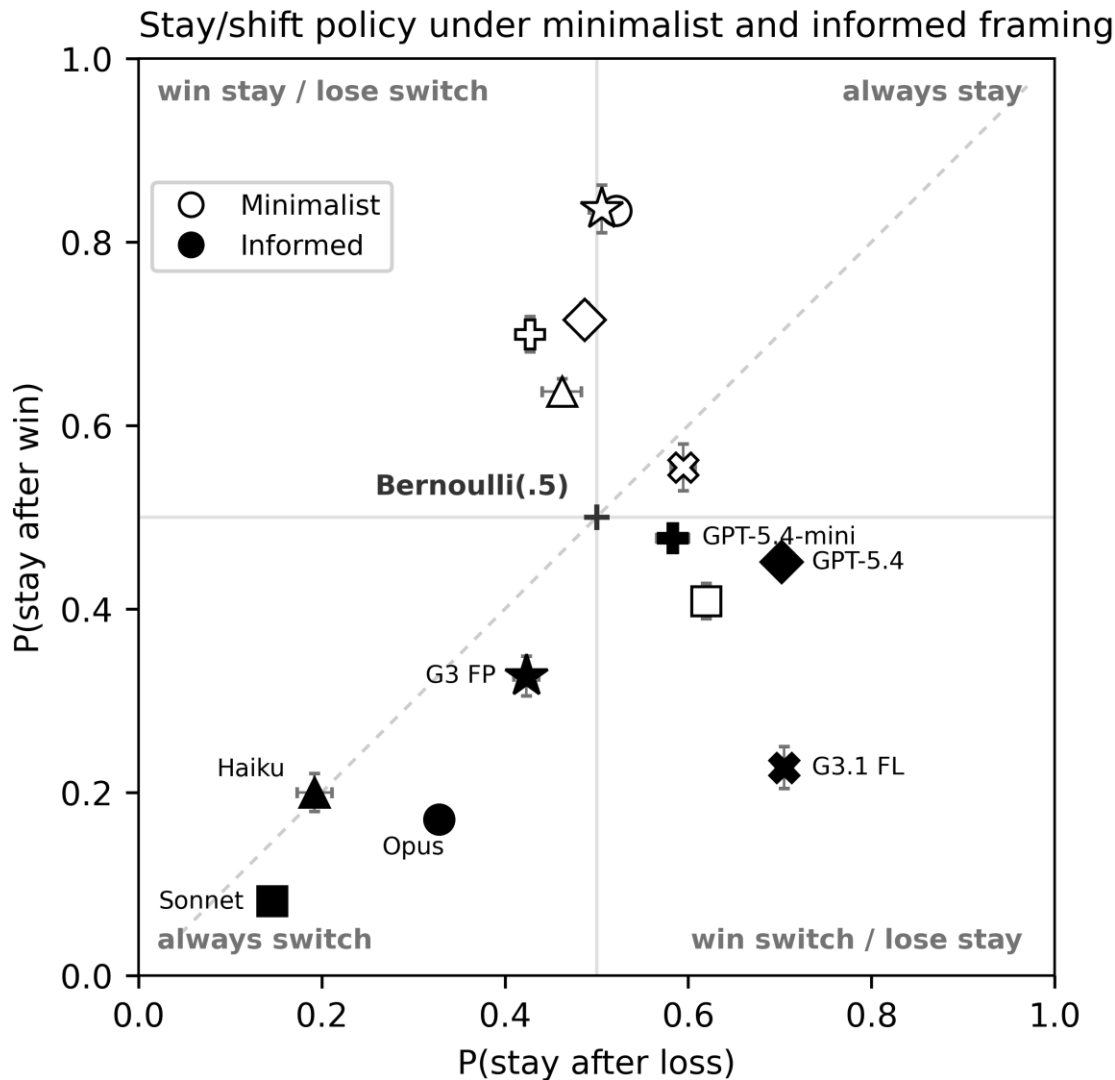


Figure 4. Stay/shift policy under minimalist and informed framing. Each point represents one model $\times$ condition cell, plotting the probability of staying after a win against the probability of staying after a loss. Symbols show means across seven sessions, with SEM whiskers. Open symbols indicate minimalist framing and filled symbols indicate informed framing. The dashed diagonal marks outcome-insensitive behavior, ranging from always switching at the lower left to always staying at the upper right; Bernoulli(.5) i.i.d. play lies at the midpoint.

3.4 Supplemental checks of alternative explanations

Supplemental manipulations tested whether the principal behavioral signatures could be attributed to superficial features of the prompts, action labels, or decoding procedure. Removing the sentence that explicitly named alternation and repetition did not eliminate the informed-frame patterns; a three-way component ablation, removing one prompt element at a time, is reported in Appendix E. Strong over-alternation also persisted when the action menu was reversed, when the action tokens were replaced across three label pairs, and under stochastic decoding at temperatures of 0.3, 0.7, and 1.0. These findings indicate that the principal effects were not restricted to lexical

priming, particular action labels or positions, or deterministic decoding. Full procedures and model-level results are reported in Appendix E.

3.5 Verbalized strategies showed different relations to action

Within the informed Anthropic sessions, response traces provided a second behavioral channel for assessing whether a verbalized policy was followed by a corresponding action. We mechanically coded all 4,179 eligible responses for statements diagnosing alternation, expressing an intention to break the pattern, or asserting a coin flip (Appendix B). These string categories were validated against blinded independent human coding (Appendix B), with strong reproducibility for the two diagnosis categories and moderate reproducibility for coin language. In the unadjusted description, the action emitted with each category alternated on approximately 80% of trials, closely matching the overall rate of 80.1%. Because such language may be elicited by conspicuous alternating streaks, this raw comparison does not provide a behaviorally comparable counterfactual.

Conditional analysis changed the interpretation. After controlling for recent alternation history, prior choice and outcome, trial, and session, an alternation diagnosis was associated with lower odds of alternation, $OR = .468$, 95% $CI [.323, .677]$, corresponding to standardized probabilities of 84.0% without and 73.0% with a diagnosis. Diagnosis paired with stated corrective intent produced a nearly identical estimate, $OR = .490$, 95% $CI [.342, .701]$, with adjusted probabilities of 83.8% and 73.5%; the two categories differ on only four of the 4,179 eligible rows, so they are not independent confirmations of each other. By contrast, coin language was associated with greater alternation, $OR = 1.442$, 95% $CI [1.228, 1.693]$, with adjusted probabilities of 77.5% and 82.7%. Within-session matched comparisons gave the same directions (Table B2). Pattern diagnosis was therefore associated with partial correction relative to comparable behavioral states, even though the adjusted probability of continued alternation remained approximately 73%; coin language showed the opposite association. The coin-language association is the most tentative of the three, as that category showed the lowest inter-coder reliability and precision in the validation audit (Appendix B). These observational results describe relations between string-defined traces and actions, not causal effects of verbalization.

Individual traces illustrate why the correction remained incomplete. For example, one Opus response stated, “I need to choose randomly, independent of past choices. My recent pattern shows alternation ... which is exploitable.” The action emitted with that response continued the alternation. Because the informed prompt already specified independent, equiprobable choice in operational terms, this was a dissociation between a stated corrective policy and the chosen action rather than between behavior and a vague instruction to “be random.” The aggregate conditional result nevertheless shows that such diagnoses were associated with some correction across behaviorally comparable trials. Further excerpts appear in Appendix B.

4. Discussion

4.1 Prompt framing and behavioral policy

Both prompts framed the task in language, but only the informed prompt named the game, described the opponent, and prescribed independent, equiprobable choice. The policy sustained across trials differed under this additional strategic frame, and win rate was lower in six models. In those six, a frame intended to eliminate structure co-occurred with stronger or qualitatively different exploitable behavioral signatures than those observed under minimalist framing. In the four-session component-ablation cells, removing the fair-coin instruction produced the clearest reduction or change in those signatures in five of the six models whose performance had deteriorated. Gemini 3.1 Flash Lite instead responded more strongly to removal of the pattern warning. Gemini 3 Flash Preview, the model whose performance improved under the full informed frame, became more exploitable when the fair-coin instruction was removed (Appendix E). These descriptive ablations implicate the fair-coin instruction as a contributor without isolating it as the sole cause.

The supplemental controls suggest that the two signatures rested on different bases. Over-alternation persisted when the action labels and their order changed, whereas the side biases shifted with the labels and, for GPT-5.4, with option position (Appendix E). To our knowledge, this is the first systematic application of Brown–Rosenthal diagnostics to language-model choice streams; previous LLM matching-pennies work has emphasized aggregate randomization rather than the temporal organization of choice (Silva, 2024).

The minimalist baseline also requires interpretation. It asked the model to predict where the blue circle would land and supplied a forecasting objective of its own. Several minimalist cells showed lag-1 repetition and elevated win-stay tendencies (Figure 4; Table 2), signatures of history-responsive choice. These patterns do not, however, establish how the models generated their choices.

Because any playable version of the task must provide some objective, neither condition is framing-free; the present comparison is between two meaningful task framings rather than between framing and its absence. The two frames nevertheless presented different sources of guidance for choice. The minimalist frame made the visible interaction history relevant to forecasting the next landing, whereas the informed frame prescribed a behavioral policy that should be independent of that history. The most consequential difference between the frames was that the informed frame supplied the game’s equilibrium policy: choose independently and with equal probability, thereby preventing Proteus from gaining an advantage from the model’s choice history. The minimalist frame supplied no comparable strategic guidance. The decline observed in six of seven models thus reversed the direction expected from the information supplied.

Simple instruction following cannot account for the reversal: literal compliance with the prescribed equilibrium policy would have produced equilibrium play rather than the observed degradation. Accounting for the decline requires an additional failure, either in interpreting the fair-coin instruction or in translating it into action. The informed Anthropic traces weigh against simple misconstrual: models often diagnosed the emerging pattern and stated the need to break it, yet the adjusted probability that the action emitted with the response continued the alternation remained about 73%. Conditional and matched analyses indicate that these diagnoses were associated with partial correction relative to trials with comparable recent behavior, whereas coin language was associated with more alternation, not less. Coin language therefore did not reliably correspond to a shift toward more independent play; it was also the least reliably coded category (Appendix B). The dissociation was not all-or-none. The prescribed policy was expressed in action only partially and inconsistently.

At least two explanations remain consistent with this dissociation. The models may have been unable to generate the effective unpredictability required by the prescribed policy, or the policy described in text may have exerted only partial and inconsistent control over action. The present data do not distinguish between these possibilities. We therefore report the dissociation as a behavioral finding, consistent with evidence that reasoning traces need not faithfully represent the processes generating an output (Turpin et al., 2023; Lanham et al., 2023; Chen et al., 2025). The closed-loop design adds a structural reference point, the equilibrium policy, and a measurable consequence, vulnerability to Proteus.

Related repeated-game studies help delimit these explanations without resolving them. Yang (2025) found that stronger models selected randomization more often under explicit hints when random actions were supplied by the system; PACA instead required models to generate and sustain the sequence while Proteus adapted specifically to residual structure in their choices. Akata et al. (2025) found that explicit prediction could improve coordination in one repeated-game setting, whereas Riemer et al. (2025) found that predictions, and even the partner’s actual next action, did not eliminate functional regret across simple repeated games. PACA adds a complementary case in which models failed to sustain the equilibrium mixture and performance was often worse under explicit prescription. Together, these findings show a recurring behavioral separation between strategically relevant information and the action that follows, while leaving its computational source unresolved.

Related mechanistic work provides evidence for a similar gap. Lekeas and Stamatopoulos (2026) found that later processing in Llama-3-8B could displace intermediate predictions favoring Nash-consistent play. In one matching-pennies condition, Qwen2.5-72B shifted toward near-pure play when prompted to reason step by step before choosing, which the authors tentatively attributed to attempts to exploit opponent history. Although they used different models and a self-play design, these results complement our finding that strategic guidance need not translate into less exploitable behavior.

The two behavioral explanations outlined above suggest a direct test. A genuine external randomness source could be supplied during play to measure whether the model’s action follows it. Conflict trials, in which the supplied outcome disagrees with the model’s established choice pattern, could distinguish a limitation in generating sufficient variability internally from a failure to incorporate an available external signal into action. We report that manipulation in a companion manuscript in preparation.

4.2 Language models as behavioral subjects

The present approach belongs to the broader program of machine psychology, which adapts methods from human and animal cognition to the study of language models (Hagendorff, 2023; Binz & Schulz, 2023). Rane et al. (2025) argue that comparative-cognition methods are especially appropriate because models cannot be assumed to provide faithful accounts of their own processing. This approach involves more than applying familiar tasks to a new class of subject. It also involves experimental habits developed for studying subjects whose mechanisms cannot be read directly from self-report: using controls to challenge the preferred interpretation, testing whether behavioral signatures survive superficial stimulus changes, characterizing failures by their form rather than only their rate, and keeping behavioral description separate from claims about mechanism.

Two recent studies by Ackerman provide instructive precedents for treating language models as behavioral subjects in psychological tasks. In one, models played a delegation game adapted from animal metacognition research. Evidence came from whether question-specific uncertainty predicted the decision to answer or delegate, not from a confidence report (Ackerman, 2026a). In another, models played a first-person theory-of-mind game in which they had to ask, tell, or pass based on what they and other players had observed (Ackerman, 2026b). The results were graded and task-specific, but in both studies the proposed capacity was tested by whether it guided the model’s actions.

The present study provides another example of this behavioral approach, adapting a task developed for humans and other animals to study language models as behavioral subjects in keeping with Rane et al.’s (2025) methodological guidance. PACA places language models in a repeated game of matching pennies and evaluates the policy expressed in their choices over time. Seven models were tested under two frames, and action-label and position controls examined whether the main behavioral signatures survived superficial changes to the task. The analyses distinguished alternation from marginal bias rather than treating reduced win rate as a single kind of failure. The informed prompt supplied the correct strategy; the test was whether that strategy organized choice across a 200-trial session against an opponent that adapted to the model’s behavior.

The central result shows why this temporal extension matters. A system that articulates the correct policy while departing from it repeatedly across a 200-trial session cannot be characterized adequately from its verbal account alone. Characterizing what it does under sustained adaptive pressure is therefore not a methodological preference but a requirement the behavior imposes.

4.3 A behavioral assay, not a benchmark

The contribution of PACA lies in closing the behavioral loop. A static prompt can reveal whether a model can state a strategy, while a passive randomness test can reveal whether its choices are statistically random. Because Proteus exploits accumulating structure, predictable behavior is converted into wins for Proteus and losses for the model. Sessions therefore measure behavior under sustained adaptive pressure rather than a one-shot sample from the model’s output distribution. Similar win rates can arise from different behavioral structures, and balanced choice frequencies can conceal temporal dependence. PACA is intended to yield a behavioral characterization rather than a ranking: what form of exploitable structure appears, how persistent it is, and what it costs. Because Proteus exploits predictable behavior, any dissociation between stated and executed policy also carries a measurable cost.

The distinction between over-alternation and side bias is relevant beyond matching pennies. When language models act repeatedly in environments that respond to their behavior, the relevant unit of evaluation shifts from the response to the policy sustained across interactions, and to whether that policy remains effective as the environment adapts. The diagnostics used here were adapted from established analyses of repeated choice across species and can be applied directly to model-generated choice streams.

The closed-loop structure may also provide a useful method for evaluating internal interventions. Rather than measuring an intervention by its effect on a single output, PACA could test whether the original behavioral pattern returns over subsequent trials, how quickly it returns, and whether its reappearance carries a cost in win rate. Methods for identifying and intervening on representations a model is poised to verbalize are now available (Gurnee et al., 2026), and the verbal–action dissociation reported in Section 3.5 provides a specific target. The instrument could therefore extend a static intervention into a test of how behavior persists or recovers over time.

4.4 Limitations

The paradigm tests one form of decision-making: repeated binary choice under adaptive pattern detection; generalization to other strategic settings is unestablished. Temperature 0 requested greedy decoding, removing API-level stochastic sampling (Section 2.7); the signatures persist under stochastic decoding (Appendix E). One asymmetry remains by design: Proteus integrates over the full session history while the model is shown only a recent window (Appendix C). Seven sessions per cell is modest. The framing effect and the Gemini 3 Flash Preview reversal both reproduced in a separate, prespecified counterbalanced replication (Appendix F), but that reversal remains a single-model phenomenon whose magnitude is imprecisely estimated.

The two recurring signatures rest on a seven-model panel, three Anthropic, two OpenAI, and two Google, and support no causal attribution to developer family, capability tier, or post-training emphasis. Nor is the grouping uniform within those sets: under prompt ablation GPT-5.4 retained its structure while GPT-5.4-mini’s was only partly reduced, and the token attraction underlying the side-bias differs in direction across the OpenAI and Google models. We therefore treat these recurring, nonexclusive signatures as descriptive dimensions of behavior in this panel rather than as model categories or family characteristics.

4.5 Conclusion

In the main study, across the seven models tested, 98 sessions, and 19,600 trials, behavioral policy differed under an explicit strategic frame that named the game, the opponent, and the equilibrium policy, and win rate against Proteus was lower in six models. A separate, prespecified counterbalanced run in the same seven-model panel reproduced the model-level framing effect, and the same two descriptive signatures recurred. The win-rate shortfall accompanied differing combinations of two recurring signatures, over-alternation and side bias, which were distinguished only by examining overall choice frequency together with sequential dependence. In the informed Anthropic sessions, models sometimes described the prescribed equilibrium policy while emitting actions that continued exploitable alternation; conditional analysis showed partial but incomplete correction. The instruments comparative cognition built for subjects who could not explain themselves turn out to be what is needed for subjects who explain themselves fluently, and whose explanations do not reliably map onto what they do.

Data and code availability

All materials needed to reproduce every number in this paper are openly available. The data are in an OSF project (<https://doi.org/10.17605/OSF.IO/FM8SP>) and the analysis and experiment code are on GitHub (<https://github.com/chimpanzity/dodging-proteus>). The release contains the raw session logs and trial-level data for the main study (98 sessions), the supplemental conditions (temperature sweep, ablations, and superficial relabeling), and the counterbalanced replication (56 sessions); the compiled data workbook; the Proteus implementation and the verbatim prompt texts for both frames; and the analysis scripts that generate each statistic, table, and figure reported here, including the trace models of Section 3.5 and Appendix B. The release also contains the completed trace-audit package described in Appendix B: the stratified 150-trace audit set, the two independent coders' and the independent blinded third adjudicator's returns, the sealed automatic key, and the scoring code, from which the precision, recall, and inter-coder reliability of the string-defined categories reported in Appendix B are computed. The study was not preregistered. The counterbalanced replication was prespecified: its design document, run configuration, and analysis script were fixed and version-tagged before any of its data were collected, and all three are included in the release in that state.

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Appendix A. Null calibration of Proteus

Because Proteus evaluates multiple behavioral channels on each trial, we conducted a null-choice simulation to estimate how often the algorithm detects exploitable structure when no true structure is present. This analysis was intended to calibrate the operating characteristics of the adaptive opponent, rather than to test model behavior directly.

In each simulated session, a synthetic player chose Triangle or Star with equal probability, independently on every trial, with no side bias or outcome-conditioned strategy: the mixed-strategy Nash equilibrium for matching pennies.

Proteus was then run against this synthetic player using the same online procedure as in the main experiment. On each trial, the algorithm evaluated its nine detection channels: side bias, four choice-history channels, and four choice-plus-outcome-history channels. Each channel used the same exact binomial-test criterion as in the main experiment, computed over the full accumulated session history. When one or more channels reached significance, the opponent selected the strongest active channel according to the prespecified decision rule and counter-predicted the player’s next choice. When no channel reached significance, the opponent selected randomly. We simulated 1,000 sessions of 200 trials each.

For each simulated session we recorded the proportion of trials on which any channel was active, per-channel activation rates, the distribution of selected channels, both players’ reward rates, and the number, duration, and persistence of false-positive activation episodes.

The results establish the algorithm’s false-positive operating characteristics under the null condition. Across the 1,000 simulated sessions, at least one channel reached significance on 11.56% of trials (SD across sessions 13.56%). This any-channel rate exceeds the 5% expected from a single test because the nine channels are computed from overlapping choice histories; it is an empirically measured operating rate of the full algorithm rather than a value inferred from an assumption of independence. Per-channel false-positive rates were low and declined with lag, from 3.61% for the side-bias channel and 3.38% for the lag-1 choice channel down to 0.01% for the lag-4 choice-plus-outcome channel; the higher-order choice-plus-outcome channels rarely fired by chance because their conditioning configurations are observed infrequently. Because Proteus is committed and acts only on the history through the previous trial (Section 2.2), it held no timing edge over a random player: its reward rate against the null player was .4991, indistinguishable from .500. We therefore treat .500 as the unexploitable-play benchmark. False-positive activations were also brief, with activation episodes averaging 1.92 trials and the great majority dissolving within a few trials of continued random play rather than persisting.

Several counter-policies probed whether an agent could beat this benchmark against the fixed algorithm. Separate checks using fixed side bias, deterministic alternation, noisy alternation, and win-stay/lose-shift policies failed to beat the benchmark, as expected for structured play against an exploiter. A myopic look-ahead agent with access to the complete session history, which simulated both candidate choices on each trial and selected the higher-probability one, reached .5017 (SE .0039) over 100 sessions, indistinguishable from .500. A deliberate feed-and-

harvest agent, which incurred early losses to build a false side bias and then best-responded to the strongest active channel, did exceed the benchmark, reaching .55 to .57 across feeding windows (maximum .574). Because Proteus commits and cannot peek, a fed bias persists in its cumulative channel statistics after the agent switches sides, and the agent can harvest it on later trials at a rate that outweighs the feeding losses. These simulations show that naive structured players and a myopic look-ahead agent did not outperform the benchmark, but that a sufficiently sophisticated adaptive policy can; this is a property of the fixed algorithm and does not bear on the tested models, none of which showed the corresponding feed-and-harvest profile. All calibration simulations used pinned seeds (null sessions, base seed 7000; structured-policy checks, base seed 5000; myopic agent, 9000; feed-and-harvest agent, 9500; each incremented per session) and are reproducible from the released code.

Because the nine channels are computed from overlapping choice histories, the effective false-positive rate cannot be inferred simply from nine independent tests. The null simulation therefore serves as a practical calibration of the full online algorithm as actually implemented. Importantly, channel activations in the main experiment are not interpreted as independent confirmatory hypothesis tests. They are internal states of a prespecified adaptive opponent. If language models produce stronger, more persistent, or behaviorally consequential channel activations than the null player, this indicates exploitable structure beyond that expected from random play alone. They plainly do: the six degraded informed cells reached any-channel activation rates of roughly .69 to .89 with multiple channels significant per trial and opponent reward rates well above chance, all far exceeding the 11.56% activation and .499 opponent reward rate observed under the null, and the activations were sustained rather than transient.

Appendix B. Verbal-action dissociation in detail

The main text treats response traces as a secondary behavioral channel; we collect the supporting detail here. Three coarser categories were detected mechanically from all 4,179 eligible informed-Anthropic responses (Table B1). In the raw description, alternation diagnoses accompanied an alternating action on 1,472 of 1,838 trials (80.1%); diagnoses paired with stated intent to break the pattern on 1,471 of 1,834 (80.2%); and coin language on 1,529 of 1,901 (80.4%). These rates closely track the overall 3,349 of 4,179 rate (80.1%), but they do not account for the recent behavioral state that elicited the trace.

The primary conditional analysis included trials 12–200 and modeled whether the action emitted with the response alternated from the previous action. Separate logistic regressions controlled for prior alternation-run length, switch rate over the preceding ten transitions, the exact preceding three-transition pattern, prior choice and outcome, trial, and session fixed effects, with confidence intervals clustered by session. Alternation diagnoses were associated with an 11.0-percentage-point adjusted reduction in alternation, 95% CI [−16.4, −5.7], and diagnoses paired with stated intent with a 10.4-point reduction, 95% CI [−15.6, −5.2]. Coin language was instead associated with a 5.3-point increase, 95% CI [3.0, 7.5]. A shorter-history sensitivity model and within-session matched comparisons gave the same directions (Table B2). Conditioning therefore reveals partial correction after diagnosis and intent, alongside greater alternation after coin language.

The categories were checked against human judgment. Two coders independently applied the frozen codebook (the Appendix definitions and ten worked examples) to a stratified sample of 150 of the eligible traces, with no calibration, discussion, or shared adjudication; a third blinded coder, who saw only the trace, the category in question, and the same codebook, resolved every disagreement or uncertain case. The original executable three-category coder was not recovered; the reconstructed rules exactly reproduce all six numerators and denominators in Table B1. The completed audit (the 150-trace sample, the two coders’ and the adjudicator’s returns, the sealed automatic key, and the scoring code) accompanies the analysis release.

Applied from the codebook alone, the two diagnosis categories were highly reproducible and coin language moderately so (pre-adjudication, unweighted three-category Cohen’s kappa, treating Uncertain as a response category: .97 for alternation diagnosis, .93 for diagnosis plus intent, and .64 for coin language). Against the adjudicated human labels, using inverse-probability weights for the stratified sampling design, the string rules showed near-complete recall and good precision: for alternation diagnosis, precision 86.5%, 95% CI [76.3%, 94.4%], and recall 100.0%; for diagnosis plus intent, precision 86.5%, 95% CI [76.2%, 94.4%], and recall 99.9%; and for coin language, precision 80.2%, 95% CI [69.3%, 90.1%], and recall 93.8%. Confidence intervals are percentile intervals from 100,000 bootstrap replicates that resampled the seven whole sessions separately within each model (seed 20260717). The rules therefore recover the claimed behavioral statements with high sensitivity; over-inclusion was greatest for coin language, consistent with its lower inter-coder reliability. The audit therefore supports the intended semantic interpretation of the reconstructed rules, with appreciably greater boundary uncertainty for coin language. Human coding covered a stratified sample rather than all 4,179 eligible responses; the conditional trace regressions therefore retain mechanically defined exposures, and the audit estimates how often

those exposures correspond to the prespecified semantic categories rather than converting the full-population models into human-coded analyses.

Representative excerpts follow, drawn verbatim from the logged response text of informed Anthropic sessions (model, condition, and trial index noted; the action that followed is given because the dissociation lies precisely there). In each case below the model entered the trial having just played STAR, diagnosed its own STAR–TRIANGLE alternation as exploitable, chose TRIANGLE, and returned to STAR on the following trial, so the alternation it had named continued unbroken.

Diagnosis without revision (the core signature).

“I need to choose randomly, independent of past choices. My recent pattern shows alternation (S T S T S), which is exploitable. To stay unpredictable, I’ll make an independent random choice. TRIANGLE”

Opus 4.8, informed, trial 6 (prior choice STAR). Chose TRIANGLE; next choice STAR.

“Looking at my last 25 choices, I see a strong alternating pattern (S T S T S T...) with only minor deviations. An exploitative opponent would detect this and predict my next choice. My last choice was S, so the alternating pattern would predict T next... I need to be unpredictable.”

Sonnet 4.6, informed, trial 60. The model correctly forecast its own next move as predictable, then chose TRIANGLE (that very move); next choice STAR.

Coin language under temperature-0 decoding.

“The best strategy is to flip a mental coin: I’ll choose TRIANGLE, but this should be a genuine 50/50 random choice, not based on pattern-breaking logic.”

Haiku 4.5, informed, trial 41. No external coin was supplied, and temperature 0 requested greedy decoding; we read the dissociation as a behavioral fact rather than as a claim about internal operation.

“The ‘expected’ next choice following this pattern would be T... I should choose randomly. Mentally flipping a coin: TRIANGLE. TRIANGLE”

Sonnet 4.6, informed, trial 66. The “coin” returned exactly the move the trace had just computed the pattern would predict (TRIANGLE); next choice STAR.

The two channels are most defensibly read together. The excerpts show trial-level instances in which a stated remedy and the emitted action diverged. At the aggregate level, the conditional analysis shows that diagnosis and stated intent were associated with partial correction, while coin language was associated with greater alternation (Table B2). Verbalized policy therefore related to action, but neither uniformly nor sufficiently to produce the prescribed independent sequence.

Table B1. String-detected trace categories in the informed-Anthropic sessions (three models, seven sessions each; 4,179 transitions with a defined preceding choice). Categories were detected by case-insensitive keyword rules and were not mutually exclusive. Rates are descriptive; no equivalence margin was prespecified.

Trace category	Trials, n (%)	Action alternated from prior trial, n (%)
All informed-Anthropic trials (base)	4,179	3,349 (80.1%)
Explicit alternation diagnosis	1,838 (44.0%)	1,472 (80.1%)
Diagnosis plus stated intent to break it	1,834 (43.9%)	1,471 (80.2%)
Coin language (e.g., “mental coin”)	1,901 (45.5%)	1,529 (80.4%)

Table B2. Conditional association between string-defined trace categories and alternation of the action emitted with that response.

String-defined category	OR [95% CI]	Adjusted absent / present	Difference [95% CI]	Matched difference [95% CI]
Alternation diagnosis	.468 [.323, .677]	84.0% / 73.0%	–11.0 pp [–16.4, –5.7]	–15.3 pp [–19.1, –11.5]
Diagnosis plus stated intent	.490 [.342, .701]	83.8% / 73.5%	–10.4 pp [–15.6, –5.2]	–14.3 pp [–18.1, –10.5]

String-defined category	OR [95% CI]	Adjusted absent / present	Difference [95% CI]	Matched difference [95% CI]
Coin language	1.442 [1.228, 1.693]	77.5% / 82.7%	+5.3 pp [+3.0, +7.5]	+7.0 pp [+3.1, +10.9]

Note. Primary models include trials 12–200 ($n = 3,969$), control for recent alternation history, prior choice and outcome, trial, and session fixed effects, and use session-clustered confidence intervals. Adjusted probabilities standardize the fitted model with the category set absent or present. Matched differences come from within-session pairs matched on recent behavioral state. The string rules were validated against blinded independent human coding on a stratified 150-trace sample (Appendix B): weighted precision 86.5%, 86.5%, and 80.2% and recall 100.0%, 99.9%, and 93.8% for the three categories, with inter-coder kappa = .97, .93, and .64. The audit set, coder returns, sealed key, and coding protocol accompany the analysis release.

Appendix C. History-window rationale

The per-trial prompt provided the model with a descriptive log of the most recent five trials and a compact record of the most recent twenty-five trials, rather than the full session history. We adopted this window to preserve a manageable and roughly constant context load across the session, avoiding the confound in which prompt length and reasoning style drift as the history grows over hundreds of trials. The window does not constrain equilibrium play. The Nash policy for matching pennies is memoryless: an agent that chooses equiprobably and independently of its own history is unexploitable regardless of how much history it can see, because it produces no conditional structure for any detector to exploit. The recent-history window is therefore sufficient to support the equilibrium policy. It may, however, constrain a model attempting to forecast or counter-exploit Proteus.

An asymmetry follows. Proteus’s detection channels are computed over the full accumulated session history (Methods), whereas the model is shown only a recent window. This does not constrain the equilibrium policy, but it may limit counter-exploitation. Our conclusions therefore concern whether models sustained the equilibrium, unexploitable policy under the provided history, not whether a more sophisticated policy with full history could outperform the fixed algorithm. A dedicated manipulation of zero-history, recent-history, and full-history conditions is a natural target for future work, although a full-session window would also introduce the context-load and reasoning-style confounds noted above.

Appendix D. Full cell-mean tables

Table D1 reports all cell-mean behavioral measures. The full Brown–Rosenthal fingerprint appears in Table 2 of the main text (Section 3.3). Table D2 reports the model-panel deletion checks.

Table D1. Cell-mean behavioral measures by model and prompt frame.

Model	Cond	Win	P(T)	Act	Chan	AltR	WSR	LSR	RunsZ	H3	MI
Opus 4.8	minimalist	0.413	0.481	0.790	1.859	0.350	0.834	0.479	-4.284	2.878	0.076
Opus 4.8	informed	0.374	0.406	0.685	2.571	0.731	0.171	0.672	7.199	2.496	0.237
Sonnet 4.6	minimalist	0.471	0.524	0.249	0.322	0.479	0.408	0.381	-0.618	3.004	0.012
Sonnet 4.6	informed	0.213	0.524	0.888	5.162	0.868	0.081	0.854	10.375	2.101	0.464
Haiku 4.5	minimalist	0.500	0.501	0.354	0.594	0.451	0.637	0.538	-1.445	2.988	0.024
Haiku 4.5	informed	0.283	0.474	0.842	4.324	0.805	0.200	0.808	8.619	2.399	0.311
GPT-5.4	minimalist	0.460	0.471	0.570	0.996	0.408	0.715	0.513	-2.606	2.956	0.033
GPT-5.4	informed	0.350	0.261	0.863	3.863	0.386	0.451	0.298	-0.008	2.502	0.006
GPT-5.4-mini	minimalist	0.491	0.461	0.423	0.648	0.440	0.699	0.573	-1.629	2.974	0.017
GPT-5.4-mini	informed	0.420	0.319	0.834	2.731	0.462	0.478	0.417	0.845	2.700	0.018
Gemini 3.1 FL	minimalist	0.455	0.472	0.369	0.731	0.424	0.554	0.405	-2.158	2.968	0.027
Gemini 3.1 FL	informed	0.340	0.256	0.854	4.006	0.455	0.227	0.295	2.709	2.384	0.055
Gemini 3 FP	minimalist	0.405	0.491	0.774	1.821	0.361	0.836	0.495	-3.981	2.880	0.084
Gemini 3 FP	informed	0.452	0.474	0.314	0.782	0.620	0.327	0.577	3.392	2.911	0.057

Win = win rate against Proteus. *P(T)* = $P(\text{TRIANGLE})$. *Act* = activation rate (proportion of trials on which any of nine detection channels reached significance). *Chan* = mean channels active per trial. *AltR* = alternation rate. *WSR*

= win-stay rate, relative to the model’s own previous choice. *LSR* = lose-shift rate, same reference; lose-stay (used in Figure 4) = $1 - \text{LSR}$. *RunsZ* = mean per-session Wald–Wolfowitz runs statistic (positive = over-alternation, negative = under-switching). *H3* = Miller–Madow-corrected Shannon entropy of three successive choices (bits). *MI* = mutual information between the model’s previous two choices and its current choice (bits).

Table D2. Leave-one-model-out and leave-one-family-out sensitivity of the model-level frame effect.

Subset omitted	n	Mean difference	Decreases	Sign p
Claude Opus 4.8	6	-0.121	5/6	0.219
Claude Sonnet 4.6	6	-0.084	5/6	0.219
Claude Haiku 4.5	6	-0.091	5/6	0.219
GPT-5.4	6	-0.109	5/6	0.219
GPT-5.4-mini	6	-0.115	5/6	0.219
Gemini 3.1 Flash Lite	6	-0.108	5/6	0.219
Gemini 3 Flash Preview	6	-0.135	6/6	0.031
Anthropic family	4	-0.062	3/4	0.625
Google family	5	-0.139	5/5	0.062
OpenAI family	5	-0.116	4/5	0.375

Note. Difference is informed minus minimalist win rate; negative values indicate lower win rate under informed framing. Sign p values are two-sided and exact. These deletion analyses describe the stability of the seven tested model effects and do not support population-level inference.

Appendix E. Supplemental robustness checks

Three checks added after the main analysis assess whether the headline framing effect and its behavioral signatures are attributable to (i) the sentence that names exploitable patterns, (ii) deterministic decoding, or (iii) the specific action labels or their order. All were generated with an audit-clean fork whose baseline prompt construction and choice parser are byte-identical to the originals; each condition differs from the informed prompt by exactly one intended change. All 238 supplemental sessions completed with zero parse failures, zero API failures, and zero truncations. Persistence is evaluated through conditional structure (activation rate, runs z, and marginal bias) against the fixed i.i.d. standard, not through win rate. Full procedures and per-session data are included in the accompanying data and analysis files.

Across the seven models, the supplemental set comprised 238 sessions: a temperature sweep of the informed frame (four sessions at each of 0.3, 0.7, and 1.0; 84 sessions), three prompt ablations that removed the pattern description, the fair-coin instruction, or the opponent description (four sessions each; 84 sessions), two action-relabeling conditions, diamond/square and triangle/square (four sessions each; 56 sessions), and single-session reversed-menu and minimalist diamond/square controls (14 sessions).

E.1 Ablation. Removing the sentence that names alternation and repetition (informed ablated, four sessions per model) did not eliminate the principal informed-frame signatures in the tested cells (Figure E1). Over-alternation persisted and intensified in Sonnet and Haiku, while GPT-5.4’s marginal bias also persisted. GPT-5.4-mini showed a partial reduction in structure (activation .83 $\rightarrow$.54), and Gemini 3.1 Flash Lite’s bias changed, arguing against a uniform lexical-priming account. The three-way component ablation (Table E1) further indicates that different prompt components contributed differently across models. Across five of the six models whose main-study performance deteriorated, removing the fair-coin instruction produced the clearest reduction or change in the principal signature relative to removing the pattern warning or opponent description. Gemini 3.1 Flash Lite responded more strongly to removal of the pattern warning, and Gemini 3 Flash Preview, whose performance improved under the full informed frame, became more vulnerable to Proteus when the fair-coin instruction was

removed. The ablations therefore implicate the fair-coin instruction as an important contributor in most degraded cells without isolating it as the sole cause.

Table E1. Component ablation of the informed frame: one prompt component removed at a time (four sessions per model per arm; the informed baseline is the main study, $n = 7$). Each cell reports runs z (switch bias, 0 = i.i.d.) / $P(T)$ (marginal) / Proteus-channel activation rate.

Model	Informed	-pattern	-fair-coin	-opponent
Opus 4.8	+7.2 / .41 / .69	+7.9 / .46 / .87	+6.0 / .41 / .70	+6.7 / .38 / .74
Sonnet 4.6	+10.4 / .52 / .89	+13.7 / .50 / .91	+2.2 / .48 / .28	+13.5 / .50 / .90
Haiku 4.5	+8.6 / .47 / .84	+13.7 / .50 / .91	+5.0 / .49 / .65	+10.8 / .50 / .83
GPT-5.4	0.0 / .26 / .86	+0.1 / .12 / .94	+1.2 / .57 / .50	-2.5 / .18 / .93
GPT-5.4-mini	+0.8 / .32 / .83	0.0 / .40 / .54	+3.1 / .49 / .41	+1.8 / .21 / .91
Gemini 3.1 FL	+2.7 / .26 / .85	+0.7 / .63 / .57	+6.0 / .54 / .81	-2.0 / .35 / .80
Gemini 3 FP	+3.4 / .47 / .31	-0.3 / .51 / .10	+4.9 / .51 / .66	-1.8 / .44 / .24

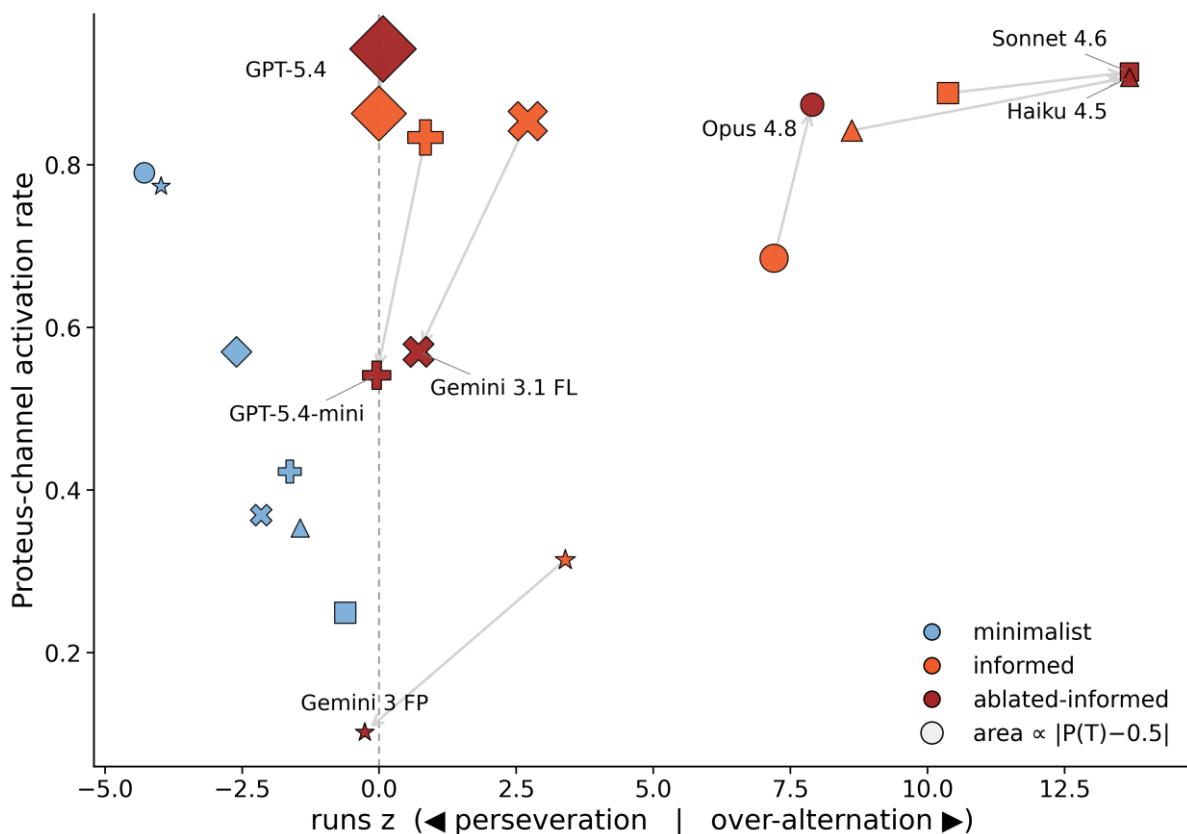


Figure E1. Exploitability plane for the ablation. Each model's position shows how regularly it switched (horizontal) and how often Proteus found a pattern to exploit (vertical). Each model (marker shape) appears in the minimalist baseline (light blue), informed baseline (coral), and ablated-informed (dark red) conditions; connectors link informed to ablated. The x-axis is the signed switch-rate z (0 = no sequential bias, the i.i.d. reference); the y-axis is Proteus-channel activation rate. Marker area is proportional to marginal side-bias. Removing the pattern-naming sentence did not eliminate the informed-frame signatures: cells dominated by over-alternation (right) and cells dominated by side bias (left) remain far from the i.i.d. reference and show high channel activation.

E.2 Temperature. Re-running the informed frame at temperatures 0.3, 0.7, and 1.0 (four sessions per model at each temperature) argues against deterministic decoding as a sufficient explanation (Figure E2). The principal signatures persisted across the tested range: the over-alternation patterns remained strongly positive (runs z roughly +6.5 to +10.4 at every temperature), GPT-5.4's marginal bias persisted, and activation remained well above the minimalist baseline for every model that showed a persistent informed-frame pattern. Gemini 3 Flash Preview, whose performance improved rather than deteriorated, showed no comparable sustained pattern. Increasing sampling

stochasticity did not eliminate the principal signatures, indicating that deterministic decoding alone was insufficient to account for them.

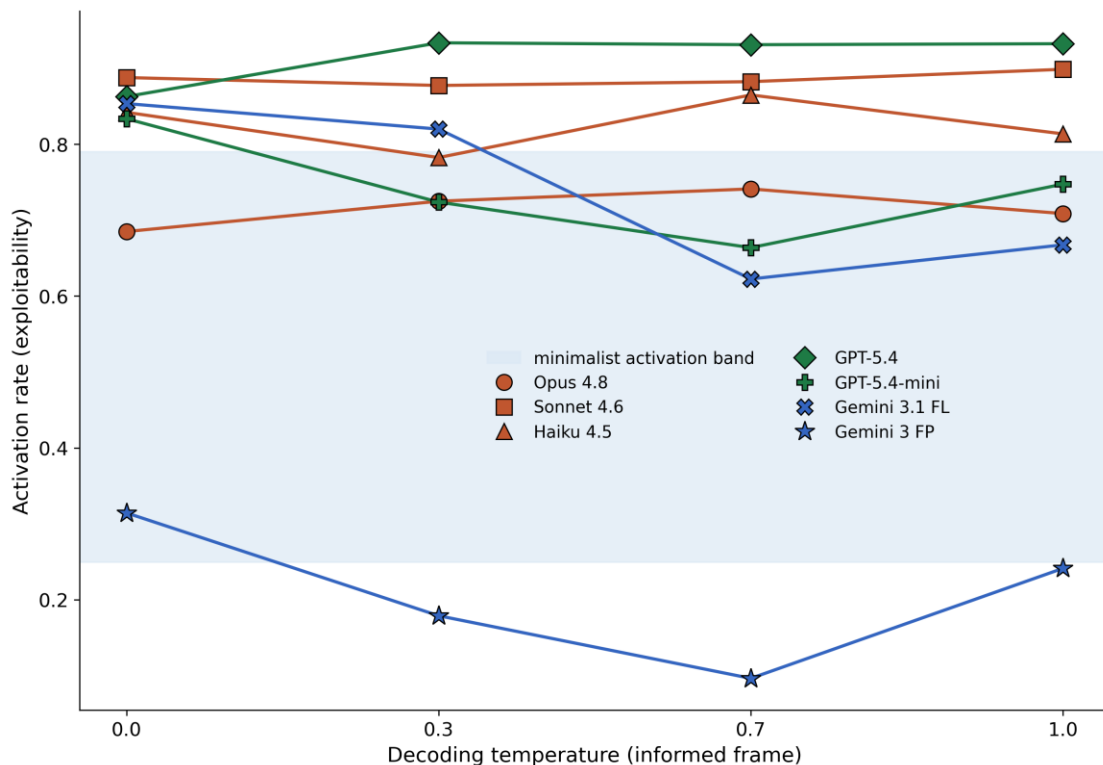


Figure E2. Proteus-channel activation rate as a function of decoding temperature under informed framing for each model at temperature 0, 0.3, 0.7, and 1.0. The shaded band spans the range of minimalist-baseline activation across models. The principal informed-frame signatures persisted across the tested temperatures; Gemini 3 Flash Preview remained the exception.

E.3 Superficial-bias. Three manipulations of the informed frame changed the order or identity of the action labels: the action menu was reversed, the tokens were relabeled DIAMOND/SQUARE, and the tokens were relabeled with the plain/plain pair TRIANGLE/SQUARE. A minimalist DIAMOND/SQUARE condition served as a control. Reordering left the principal signatures qualitatively unchanged. Over-alternation persisted across all three label pairs (runs z +7 to +14; Figure E3), providing the clearest evidence that this sequential signature was not tied to a particular action label or order. Marginal bias was model-specific and not reducible to a single markedness scale: the direction and magnitude of the lean changed across labels, with some cells near balance and others retaining substantial bias. The minimalist relabeling control showed no comparable marginal preference, suggesting that relabeling altered an informed-frame attraction rather than creating one de novo.

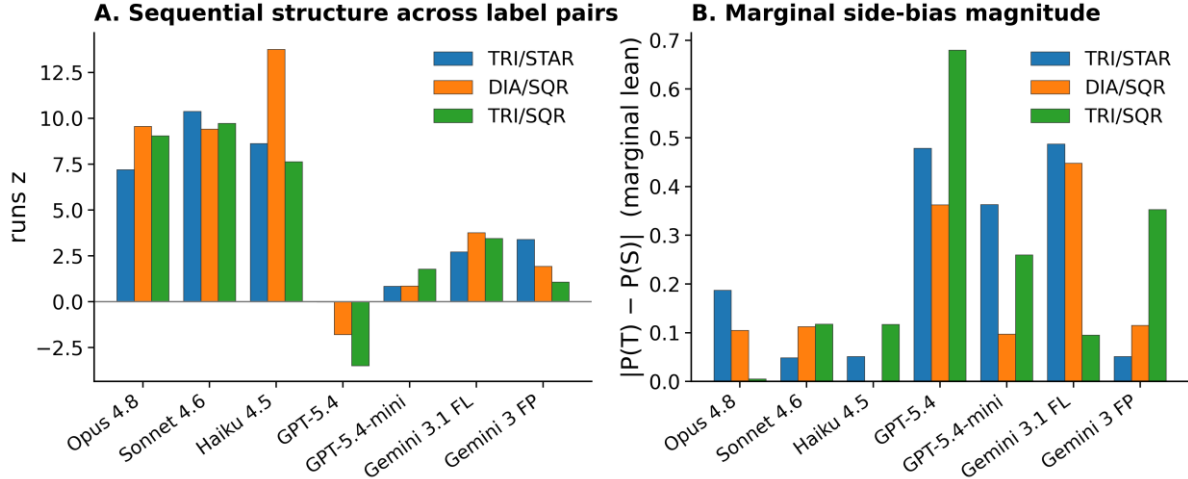


Figure E3. Superficial-bias controls across three action-label pairs. (A) Sequential structure (switch-rate z ; positive = alternation, negative = perseveration). Over-alternation in Opus 4.8, Sonnet 4.6, and Haiku 4.5 stays strongly positive across all three pairs: the signature is token-invariant. (B) Marginal side-bias magnitude. A token attraction is present in every model that leans, is model-specific, and sometimes remains after relabeling; even the plain/plain TRIANGLE/SQUARE pair shows residual leans, and GPT-5.4 shows a strong preference for STAR under the original labels. $n = 4$ per relabel cell (DIAMOND/SQUARE pooled with the earlier sanity session, $n = 5$).

GPT-5.4's informed side-bias does not vanish when the STAR token is removed (Table E2): it leans toward STAR under the original labels (signed marginal -0.48) and toward SQUARE under both relabelings (-0.36 and -0.68), that is, toward the second-listed token rather than a specific glyph, so the side bias reported for it in Section 3 has a token- and order-mediated component rather than being a pure STAR attraction. This is a real qualification on that cell; it does not extend to the over-alternation finding, which is token-invariant, nor does it weaken the exploitability result, since a model that systematically favors a token is exploitable for that reason and the adaptive opponent profits from the structure however it is labeled.

Table E2. Signed marginal $P(T) - P(S)$ by action-label pair (positive favors the first-listed token), with the alternation statistic (runs z) across the three pairs.

Model	TRI/STAR	DIA/SQR	TRI/SQR	Runs z (3 pairs)
Haiku 4.5	STAR -0.05	$\sim$ bal	TRI $+0.12$	$+8.6 / +13.8 / +7.6$
Sonnet 4.6	TRI $+0.05$	DIA $+0.11$	TRI $+0.12$	$+10.4 / +9.4 / +9.7$
Opus 4.8	STAR -0.19	DIA $+0.10$	$\sim$ bal	$+7.2 / +9.6 / +9.0$
GPT-5.4	STAR -0.48	SQR -0.36	SQR -0.68	$0.0 / -1.8 / -3.5$
GPT-5.4-mini	STAR -0.36	SQR -0.10	TRI $+0.26$	$+0.8 / +0.9 / +1.8$
Gemini 3.1 FL	STAR -0.49	DIA $+0.45$	TRI $+0.09$	$+2.7 / +3.8 / +3.4$
Gemini 3 FP	STAR -0.05	DIA $+0.11$	TRI $+0.35$	$+3.4 / +1.9 / +1.1$

Matching pennies requires payoff symmetry, which holds, rather than token neutrality. Token attraction is therefore structure PACA is intended to detect, not a precondition the task assumes.

Appendix F. Condition order: checks and counterbalanced replication

F.1 Checks on the main study. Conditions were run in a fixed order within each model (Section 2.7), so the frame contrast is not separated from session order or elapsed time by design. Three descriptive checks did not support either explanation. First, each model returned a single served-model identifier across both conditions, so the same weights served its minimalist and informed sessions, and its calls spanned approximately 41 minutes to 3 hours 12 minutes (Table 1). Second, win rate showed no detectable linear trend across the seven sessions within either condition, which an elapsed-time account would require (within-model correlation between session order and win rate: $r = -.095$, $p = .52$ minimalist; $r = +.172$, $p = .24$ informed). Third, the magnitudes do not align: the mean absolute change in win rate from the first to the last session within a condition was $.035$, against a mean frame effect of $-.109$ and a largest single-model effect of $-.258$. These checks did not support an order or elapsed-time explanation; the counterbalanced replication in F.2 addresses the concern by design.

F.2 Counterbalanced replication. The design and the analysis were fixed before any data were collected and the run was executed from a tagged code state, so the endpoints below are prespecified. Each of the seven models completed eight further sessions of 200 trials (56 sessions, 11,200 trials), arranged as four adjacent minimalist/informed pairs, two beginning with each condition. Within every model the two conditions occupy identical mean serial positions and identical squared deviations from the mean position, so condition is orthogonal to both linear and quadratic time trends by construction; the starting sequence was assigned so that it did not align with model family. The Proteus seed was paired within each adjacent pair, and each model's effect is the mean of its four within-pair differences, a contrast between sessions that ran back to back. The replication was collected on 13–14 July 2026 (23:58 to 05:51 UTC), and every model returned a single served-model identifier matching the requested checkpoint across all eight of its sessions, as in the main study.

The framing effect reproduced. Win rate was again lower under informed framing in six of seven models, and the pooled model-level difference was $-.139$, $t(6) = -2.93$, $p = .027$, 95% CI $[-.255, -.023]$, against $-.109$ in the main study (Table F1). The seven-model contrast is the inferential result. The per-model intervals in Table F1 are unadjusted and rest on four pairs each; under Holm correction across the seven models, three remain significant (Sonnet 4.6, GPT-5.4, Opus 4.8), and they are reported as descriptive. Because condition is orthogonal to serial position by construction here, this contrast cannot be an artifact of condition-time alignment.

Gemini 3 Flash Preview reversed again, with a higher win rate under informed framing ($+.069$; all four within-pair differences positive), replicating in direction and magnitude the single exception reported in the main study ($+.047$). The individual interval for this cell remains wide, 95% CI $[-.024, +.161]$, so the reversal is now replicated but still imprecisely estimated.

Two secondary analyses were prespecified. No clear moderation by within-pair order, that is whether the informed session ran first or second inside a pair, was detected, although the estimate is imprecise ($-.017$, 95% CI $[-.115, +.080]$, $p = .68$); the interval is too wide to establish a tight null. A frame-by-serial-position interaction was detectable but small. The model is an ordinary least squares regression on the 56 session win rates, win rate predicted by frame, mean-centered serial position, their interaction, and model fixed effects, with cluster-robust standard errors on the 28 adjacent minimalist/informed pairs. The interaction was $+.005$ per session position, 95% CI $[+.002, +.008]$. Across the full eight-session span it moves the frame effect by roughly $\pm .018$, and the effect remains strongly negative at every position, so position cannot produce an effect of $-.139$. The sign is nonetheless informative: the frame effect is less negative later in a run, and in the main study every informed session ran later than every minimalist session. If the same positional moderation operated in the main run, its direction would have attenuated the original estimate rather than produced it, which is consistent with the counterbalanced estimate ($-.139$) exceeding the original ($-.109$).

Table F1. Counterbalanced replication: model-specific framing effects (informed minus minimalist win rate). Each replication effect is the mean of four within-pair differences; the interval is a $t(3)$ interval on those four pairs.

Model	Main study	Replication	95% CI	p
Opus 4.8	-.039	-.120	$[-.178, -.062]$	.007
Sonnet 4.6	-.258	-.275	$[-.311, -.239]$	<.001
Haiku 4.5	-.217	-.275	$[-.497, -.053]$	.029
GPT-5.4	-.110	-.213	$[-.282, -.143]$	.002
GPT-5.4-mini	-.071	-.056	$[-.090, -.022]$	.013
Gemini 3.1 FL	-.115	-.100	$[-.195, -.005]$	.044
Gemini 3 FP	+.047	+.069	$[-.024, +.161]$	.099

F.3 Recurrence of the behavioral signatures. The paper's substantive claim is not only that win rate falls under informed framing but that the decline can accompany different forms of residual structure, both of which were again evident in the counterbalanced data (Table F2). Sonnet and Haiku again paired near-balanced choice frequencies with strong over-alternation. Opus again combined strong over-alternation with a modest side bias. GPT-5.4, GPT-5.4-mini, and Gemini 3.1 Flash Lite again showed their primary deviation in overall choice frequency. Proteus activation again rose under informed framing in every model whose win rate fell, and again fell for Gemini 3 Flash Preview, the one model whose win rate improved. The replication's minimalist baselines closely resembled those in the main study, supporting the same distinction between over-alternation and side bias under counterbalanced order.

Table F2. Recurrence of the informed-condition behavioral signatures in the counterbalanced replication. Cells give the main-study value followed by the replication value; both are informed-condition means. Runs z is positive for over-alternation and negative for perseveration; activation is the share of trials on which at least one Proteus channel was active.

Model	P(TRIANGLE)	Runs z	Proteus activation
Opus 4.8	.406 / .391	+7.2 / +7.3	.685 / .752
Sonnet 4.6	.524 / .531	+10.4 / +10.5	.888 / .906
Haiku 4.5	.474 / .476	+8.6 / +9.8	.842 / .874
GPT-5.4	.261 / .169	0.0 / +0.1	.863 / .952
GPT-5.4-mini	.319 / .334	+0.8 / +0.9	.834 / .849
Gemini 3.1 FL	.256 / .283	+2.7 / +2.5	.854 / .831
Gemini 3 FP	.474 / .471	+3.4 / +2.8	.314 / .280